Panskura Banamali College (Autonomous)

B.Sc.(H) 2nd Semester Examination , 2022

Subject: BCA Paper code: CC3T

Paper Name : Data Structure

F.M. : 40 Time : 2 Hours

---

1. Answer any four questions: 4x10=40

a. Explain the pointer with example.
Write an algorithm for selection sort.
Discuss with help of an example. 2+4+4

b. What is Stack?
Why it is known as LIFO?
Write an algorithm using PUSH and POP.
How queues are represented in memory?
Write their applications. 1+1+3+3+2

c. What do you mean by Link list?
Write an algorithm to insert a node in Singly LinkedList.
What is problem analysis? Explain. 2+3+2+3

d. Write an algorithm for Binary Search.
How it is better from Linear Search?
Explain Inorder, Preorder and Postorder Traversal operation on Binary tree with example. 3+2+5

e. What do you mean by data and data structure?
What is need of algorithm complexity and how it is evaluated?
Describe any notation used to represent algorithm complexity. 2+3+5

f. What are the various non-linear data structures?
Give example of Max heap and Min heap. 4+6

***